

Accessible tools for deep learning in bioimage analysis

Showcase of DL tools a Python-capable microscopy user can use today

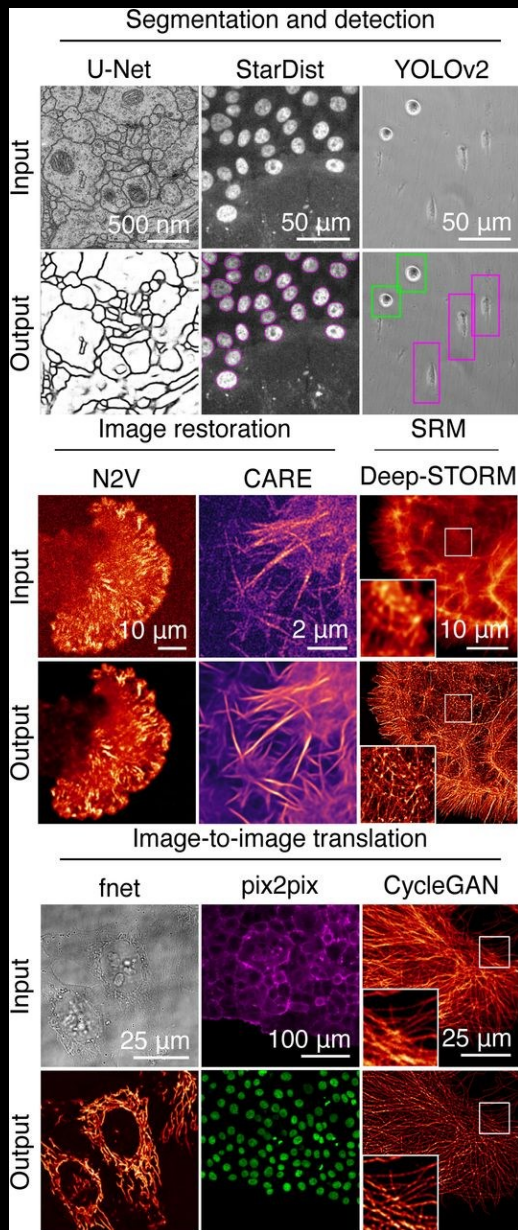
Zuzana Čočková - BioImage analyst at IMCF



Imaging Methods
Core Facility



Faculty of Science
Charles University



DL technologies are powerful

- but can be hard to access for non-experts

👥 Community efforts are lowering the barrier



MODELS EXIST
pretrained
publicly available

TOOLS EXIST
free to use
low-code setup
workflow integration

ZeroCostDL4Mic: an open platform to use Deep-Learning in Microscopy
doi: <https://doi.org/10.1101/2020.03.20.000133>

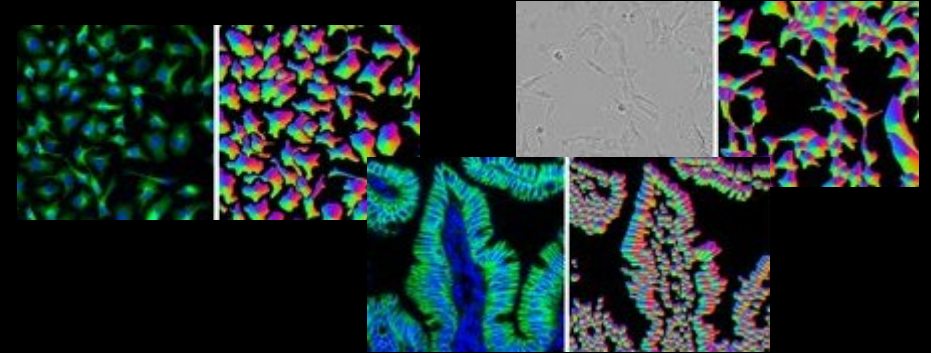


Ready-to-use segmentation solutions

- Cellpose



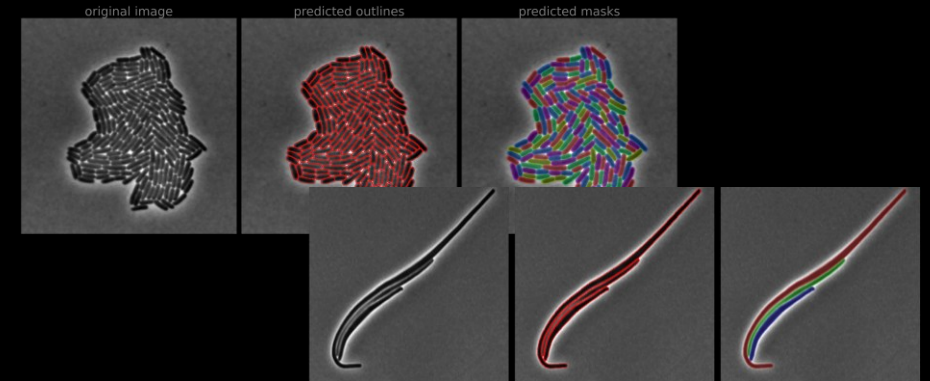
- Generalist cell segmentation algorithm
- Works on many modalities (bf, fluorescence, histology)
- Pre-trained models, re-training/fine-tuning possible



- Omnipose



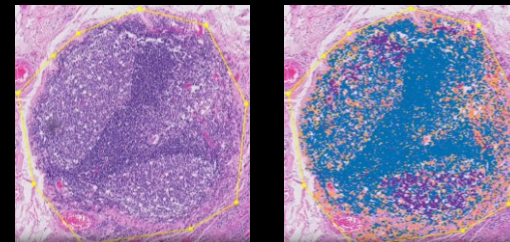
- Extension over Cellpose
- Better handling of elongated cells, irregular morphology, dense bacterial images



- Classpose



- Extension over Cellpose
- Semantic classification in whole slide images
- Available as QuPath plugin

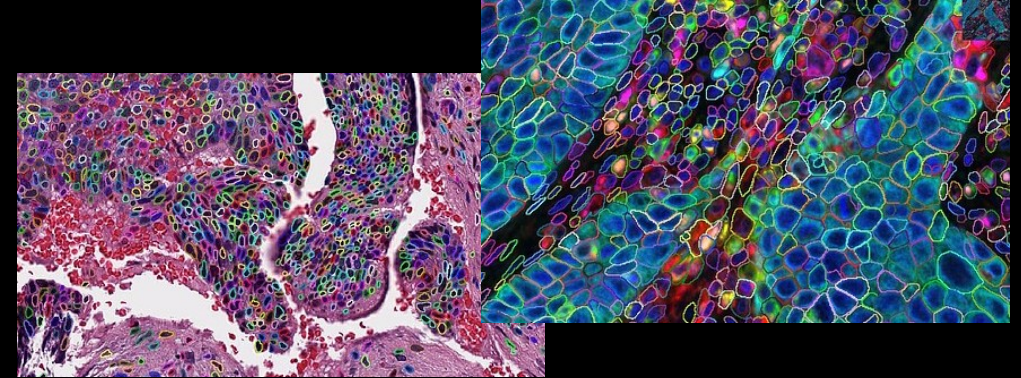


Ready-to-use segmentation solutions

- **InstanSeg**



- for cell & nuclei segmentation
- fluorescence, histology
- works with more channels at the same time



- **DeepCell**

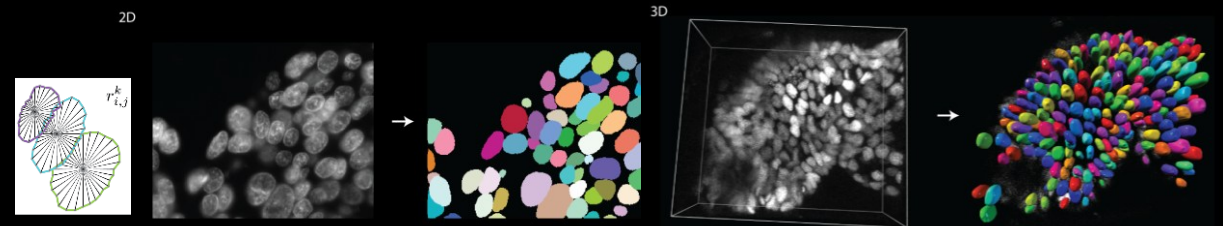


- for cell & nuclei segmentation, tracking

- **StarDist**



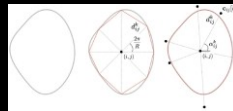
- represents objects as star-convex polygons
- excellent for roundish objects (e.g. nuclei)



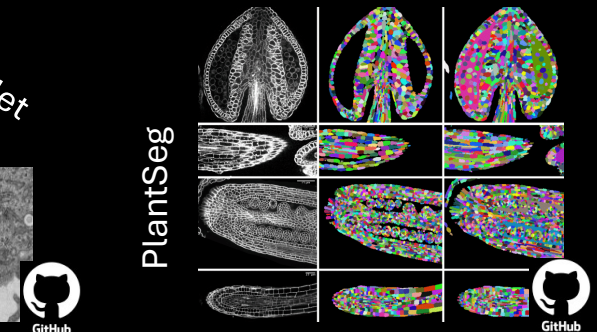
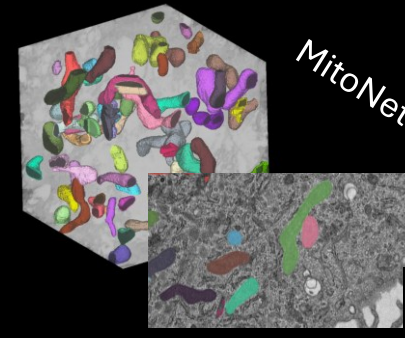
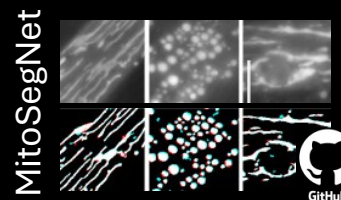
- **SplineDist**



- extension over StarDist
- represents objects as parametric spline curves



- Domain-specific segmentation...



Ready-to-use segmentation solutions

Segment Anything Model (SAM) - AI model developed by Meta AI

- <https://sam3ai.com/model/>
- can identify and segment any object in an image, is promptable



Multiple adaptations for microscopy (napari plugins):

- micro_sam (Segment Anything For Microscopy) 
- napari-sam 
- napari-segment-anything 
- napari-segment-everything 
- napari-sam3-assistant 
- napari-SAM4IS 

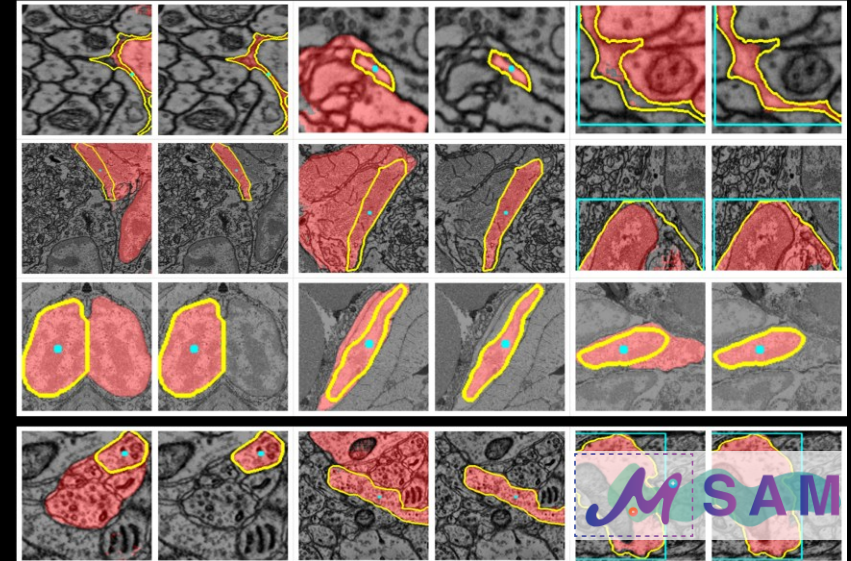
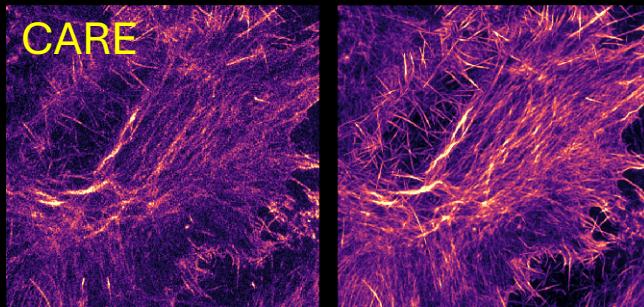


Image restoration

CAREamics  - a PyTorch library simplifying the use of supervised and self-supervised DL image restoration algorithms

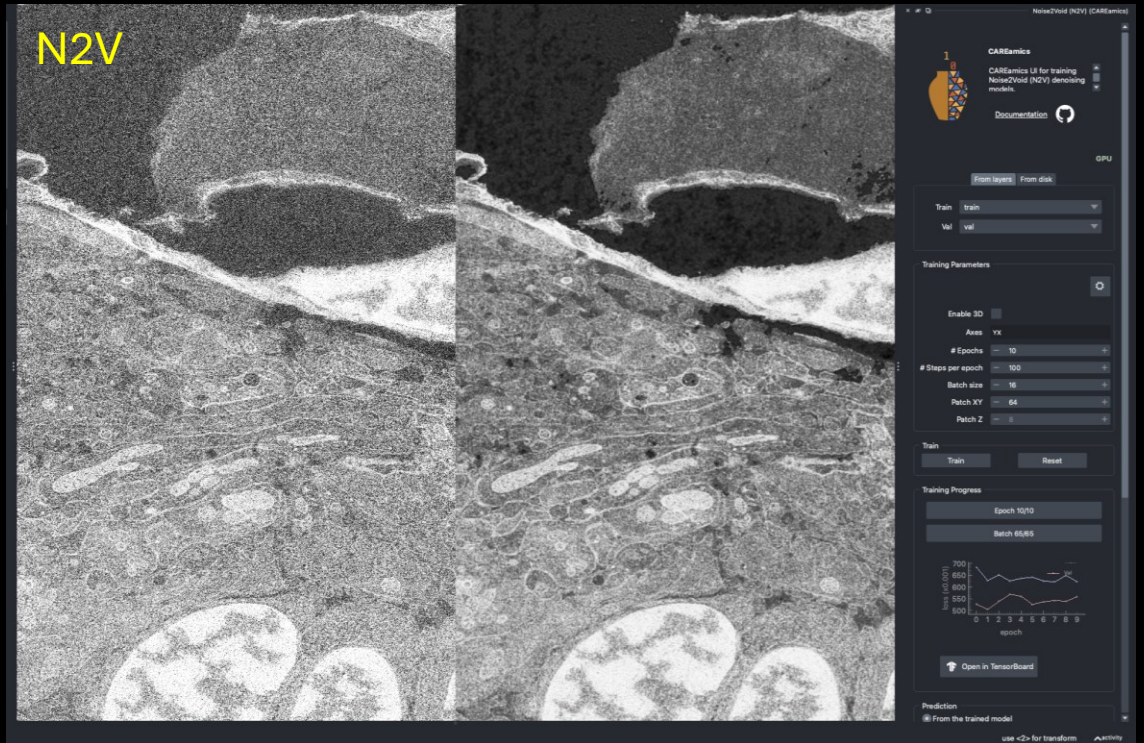
One library, many restoration algorithms

- Content-aware Image Restoration (CARE)
- Noise2Noise (N2N)
- Noise2Void (N2V, N2V2, structN2V)



low-res

high-res

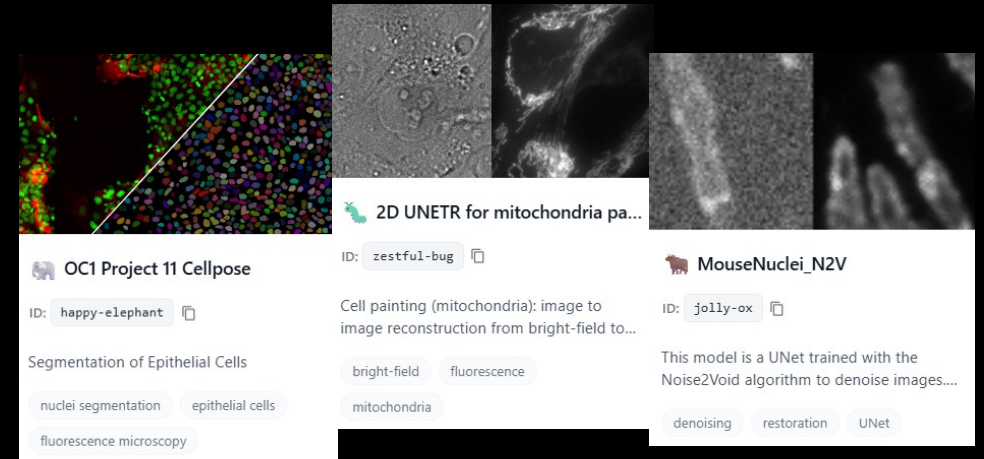


Model hubs (repositories)



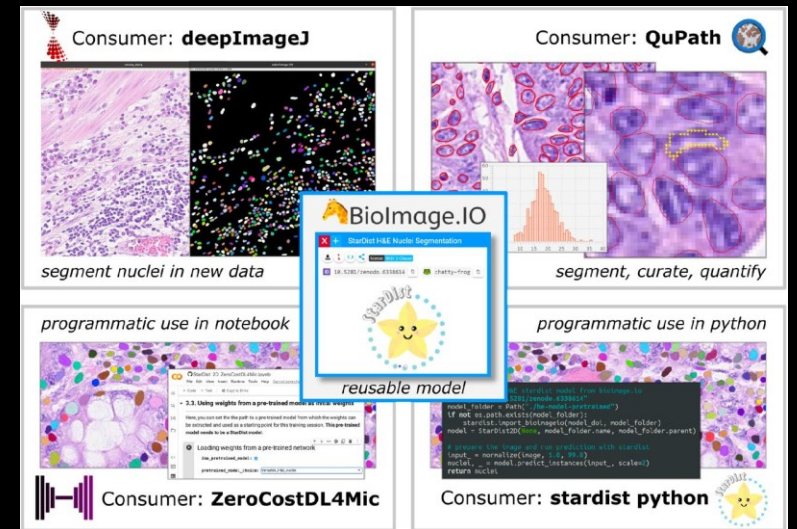
BiolImage Model Zoo

- community-driven
- standardized models
- interoperable formats - ensures models work across partner tools
- covers a wide range of tasks (segmentation, detection, restoration, classification, image-to-image translation)

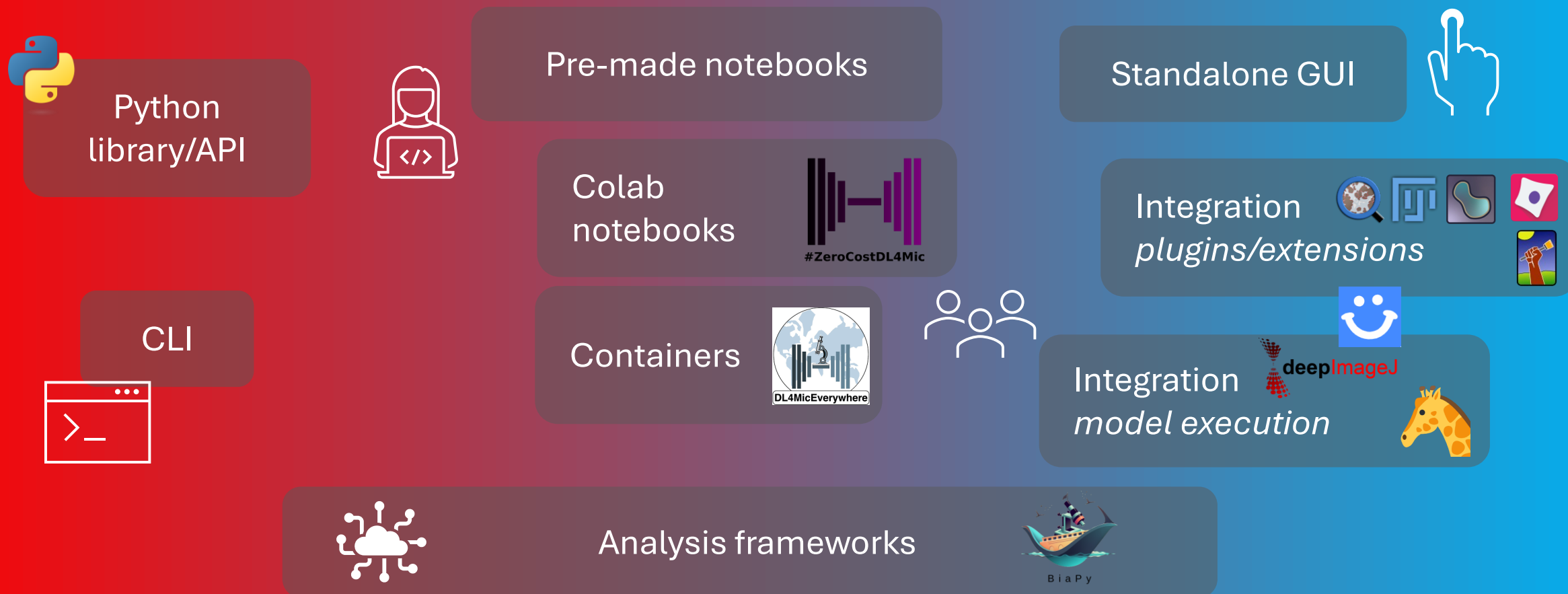


Hugging Face

- broader ML ecosystem
- growing number of biomedical and microscopy models



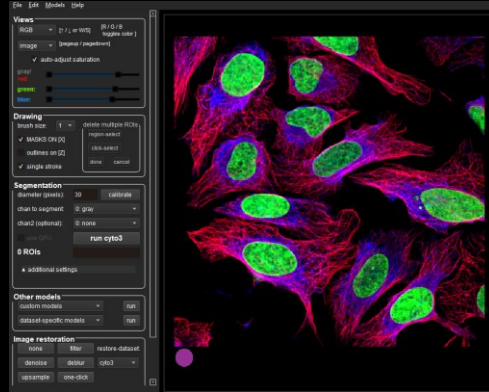
Interoperability - ways to use DL techniques



programming/technical skills requirements

Example: Cellpose ecosystem

1. via GUI



2. via Python code

```
from cellpose import models, io

image = io.imread('path/to/image')

model = models.Cellpose(model_type='cyto')

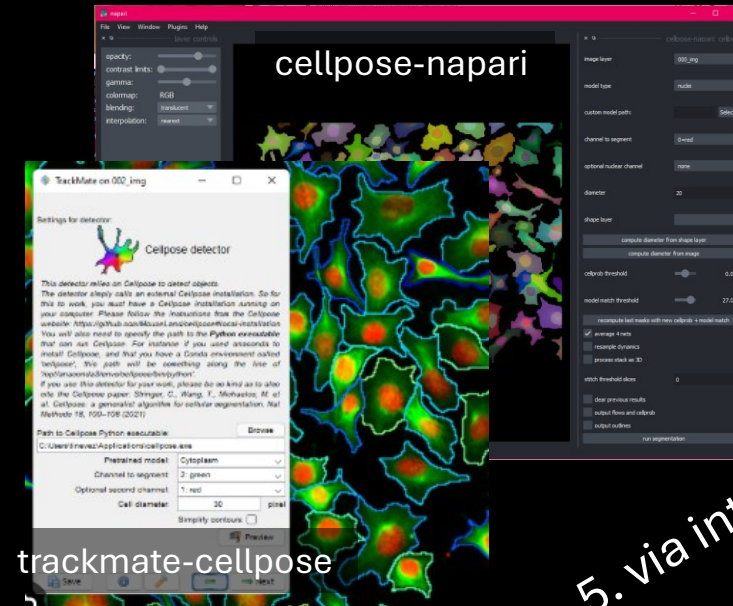
masks, flows, styles, diams = model.eval(
    image,
    diameter=30,
    channels=[0, 0]
)
```

3. via CLI

```
C:\Users\zuzka> cellpose --dir ./images --pretrained_model cyto3 --diameter 30 --chan 0 --save_png
```

4. via pre-made notebooks, Colab notebooks or containers

Cellpose (2D and 3D)	here	Instance segmentation (Cells or Nuclei)	Fully supported	05/09/23 ✓ working (IH)	here ZeroCostDL4Mic	
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Integrations may have limited functionality

- limited training
- limited batch processing
- only 2D support
- reduced parameter control

5. via integrations

BMZ model

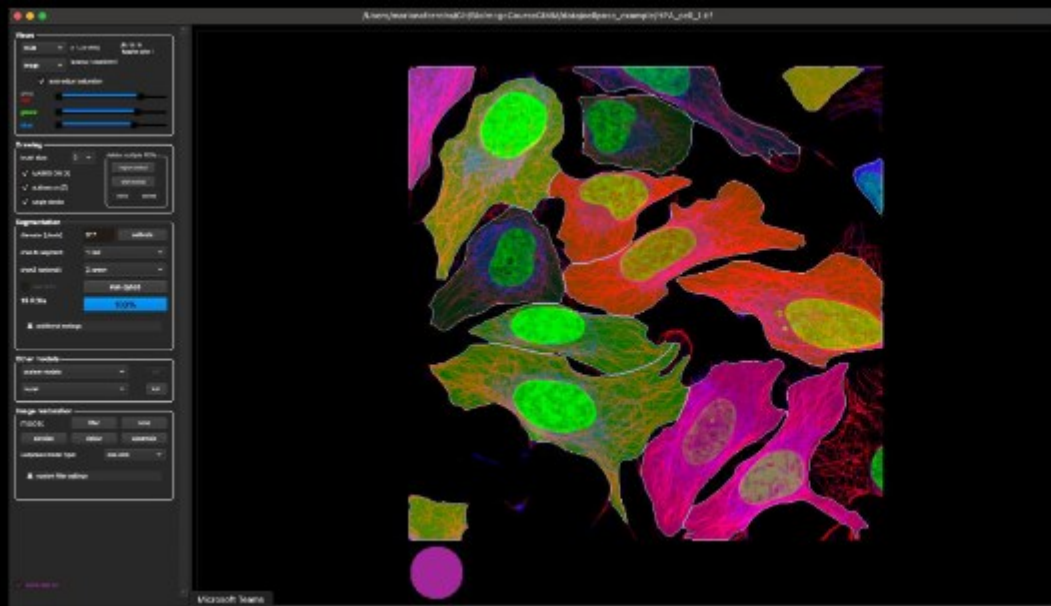


DL use case - Cellpose



Hands-on
program 

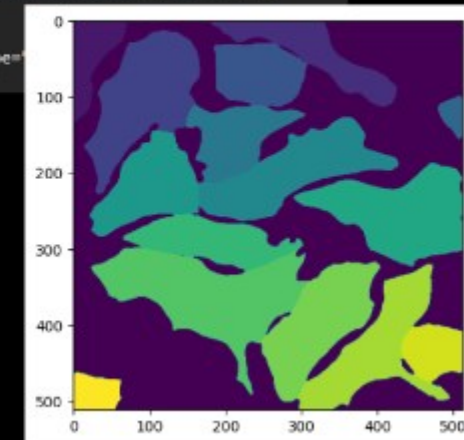
- Notebook #01
 - approach A: built-in GUI
 - approach B: via Python code (local notebook and Google Colab)



```
from cellpose import io, models
import matplotlib.pyplot as plt

# import the image
img = io.imread('../data/cellpose_example/HPA_cell_1.tif')

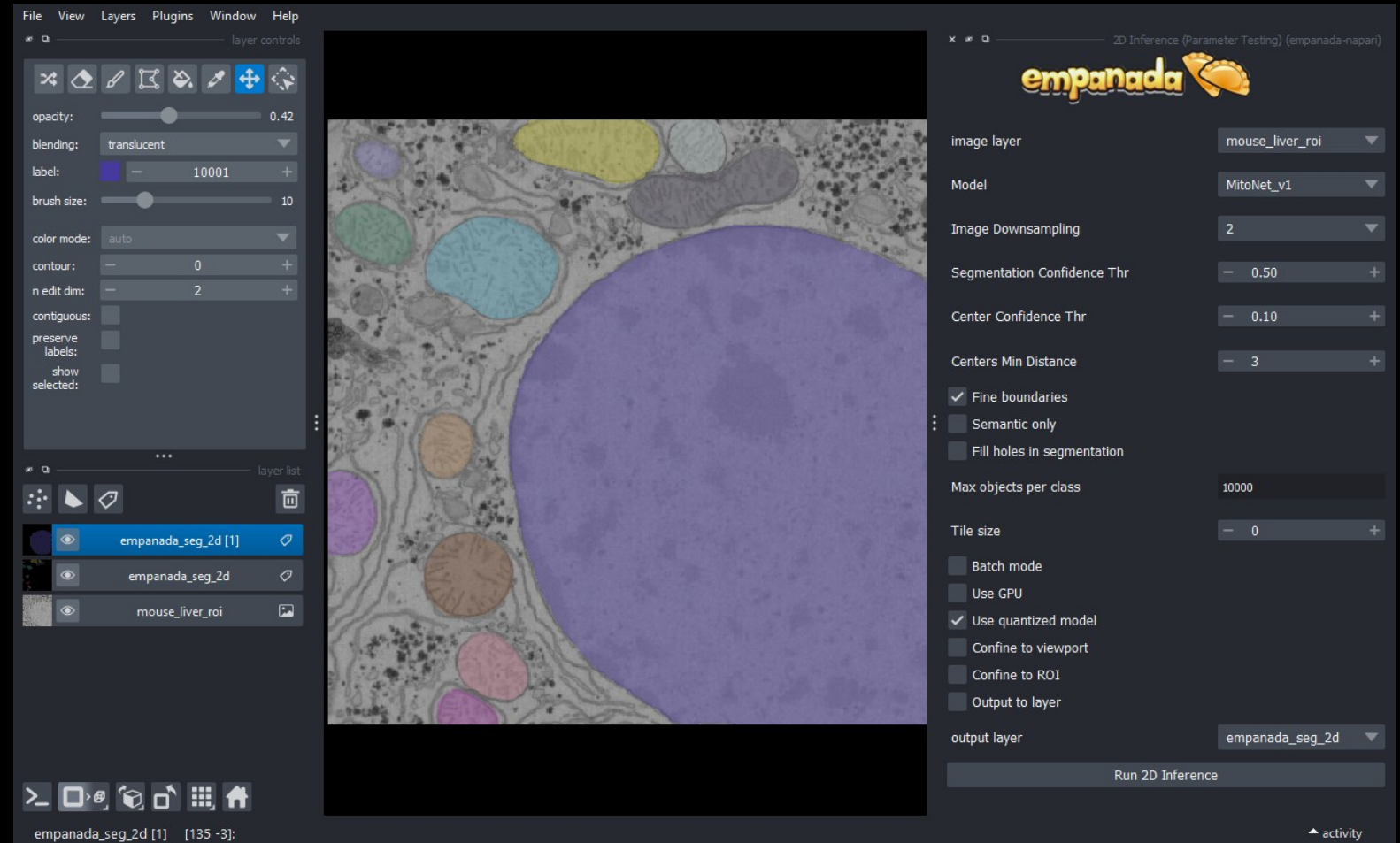
# Initialize the model
model = models.Cellpose(model_type='cyto')
```



DL use case - empanada

Hands-on
program 

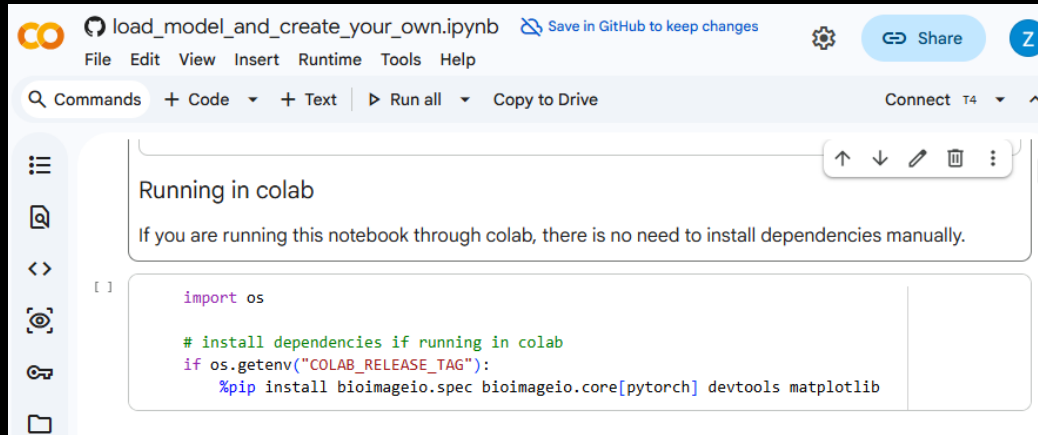
- Notebook #02
 - approach: napari



DL use case - BioImage Model Zoo

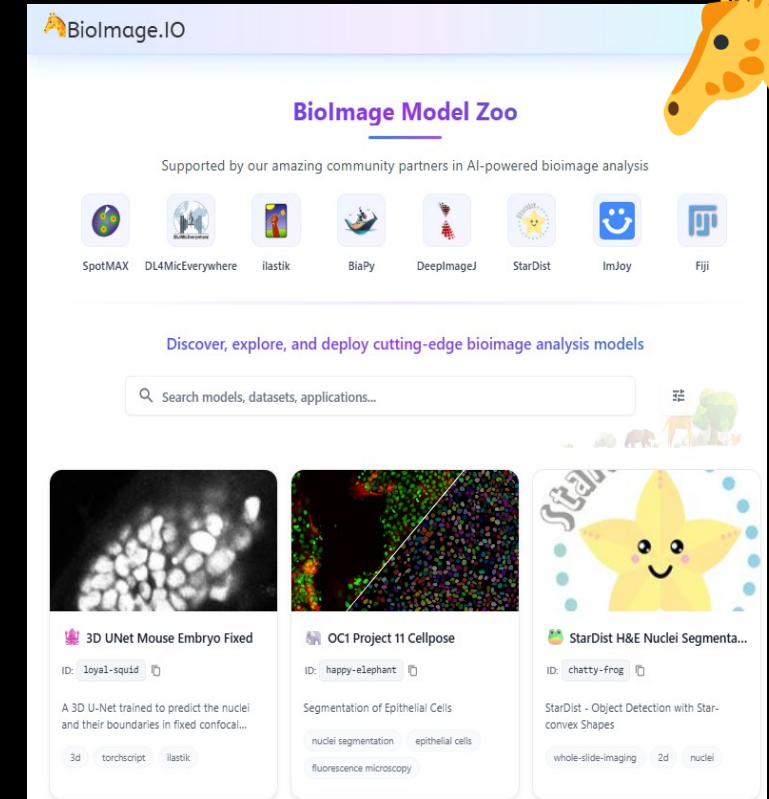
Hands-on
program 

- Notebook #03
 - test via BioEngine
 - use via Python code in Google Colab



```
import os

# install dependencies if running in colab
if os.getenv("COLAB_RELEASE_TAG"):
    %pip install bioimageio.spec bioimageio.core[pytorch] devtools matplotlib
```



ambitious-sloth 

chatty-frog 

happy-elephant 

loyal-squid 

wild-pig 

trustworthy-llama 

serious-turtle 